

Generating 3D small binding molecules using shape-conditioned diffusion models with guidance

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Drug development is a critical but notoriously resource- and time-consuming process. Traditional methods, such as high-throughput screening, rely on opportunistic trial and error and cannot ensure optimal precision design. To overcome these challenges, generative artificial intelligence methods have emerged to directly design molecules with desired properties. Here we develop a generative artificial intelligence method DiffSMol for drug discovery that generates 3D small binding molecules based on known ligand shapes. DiffSMol encapsulates ligand shape details within pretrained, expressive shape embeddings and generates binding molecules through a diffusion model. DiffSMol further modifies the generated 3D structures iteratively using shape guidance to better resemble ligand shapes, and protein pocket guidance to optimize binding affinities. We show that DiffSMol outperforms state-of-the-art methods on benchmark datasets. When generating binding molecules resembling ligand shapes, DiffSMol with shape guidance achieves a success rate 61.4%, substantially outperforming the best baseline (11.2%), meanwhile producing molecules with de novo graph structures. DiffSMol with pocket guidance also outperforms the best baseline in binding affinities by 13.2%, and even by 17.7% when combined with shape guidance. Case studies for two critical drug targets demonstrate very favourable physicochemical and pharmacokinetic properties of generated molecules, highlighting the potential of DiffSMol in developing promising drug candidates.

Drug development is a critical but notoriously resource- and time-consuming process¹. It typically takes 10–15 years and US\$1–1.6 billion to fully develop a successful drug². To expedite the process and improve cost efficiency, tremendous research efforts have been dedicated to developing computational methods to facilitate drug development³. Existing computational methods to design potential drug candidates (for example, small molecules) could be categorized into ligand-based drug design (LBDD)⁴ and structure-based drug design

(SBDD)⁵, which search molecule libraries to identify those resembling known ligands or binding to known binding sites of protein targets, respectively. Although promising, the opportunistic trial-and-error paradigm underpinning LBDD and SBDD is often confined by the limited scale of molecule libraries and cannot ensure optimal precision design⁶. Thus, the outcomes are highly subjective to the knowledge and experience of the domain experts conducting the experiments, which also limits the scalability and automation of rapid drug design

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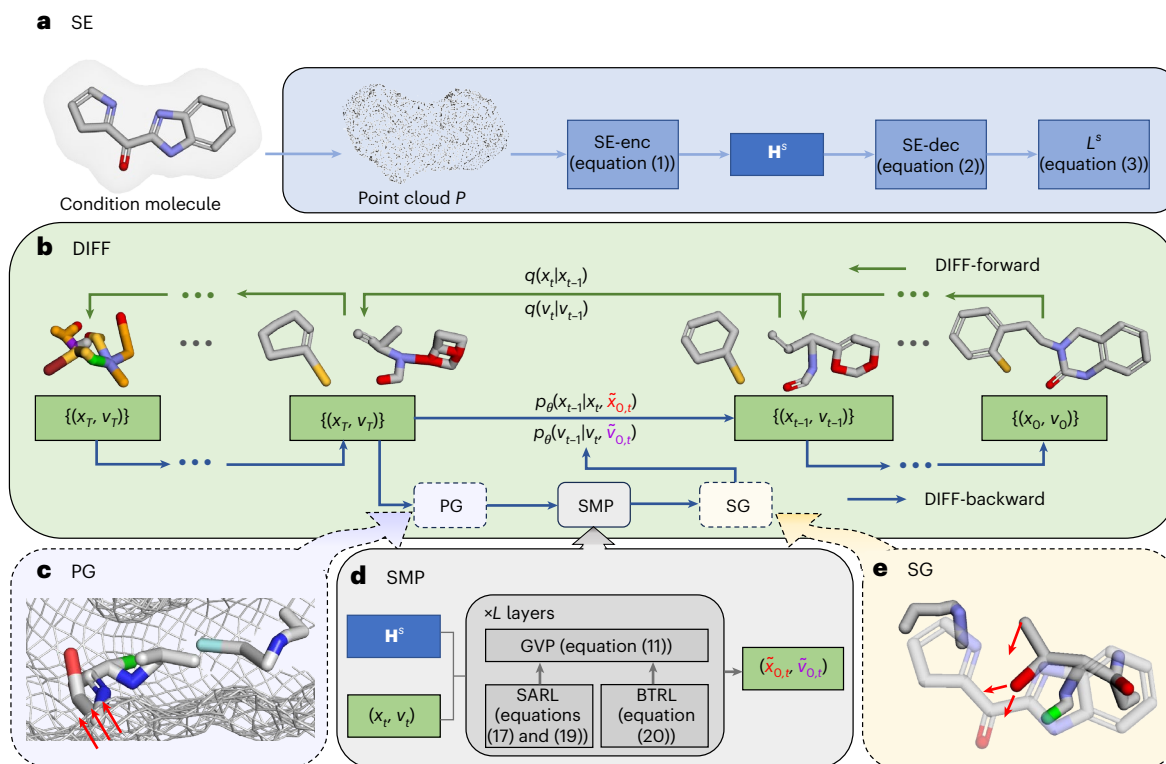


Fig. 1 | Model overview of DiffSMol. **a**, Shape embedding module SE. DiffSMol uses a shape embedding module SE to map the 3D molecule surface shapes $\mathcal{P}$ into shape embeddings $\mathbf{H}^s$. SE uses an encoder SE-enc to map $\mathcal{P}$ into $\mathbf{H}^s$, and a decoder SE-dec to optimize $\mathbf{H}^s$ with loss $\mathcal{L}^s$. **b**, Shape-conditioned molecule diffusion model DIFF. DiffSMol uses DIFF to generate molecules conditioned on $\mathbf{H}^s$. DIFF includes a forward diffusion process, denoted as DIFF-forward, which gradually adds noises step by step to the atom positions and features $\{(x_t, v_t)\}$ at step t . DIFF uses a backward generative process, denoted as DIFF-backward, to remove the noises in the noisy molecules. DIFF generates a 3D molecule by first sampling noisy atom positions and features $\{(x_T, v_T)\}$ at step T and then removing

the noises step by step until t reaches 1. **c**, Pocket guidance PG. During the generation, DiffSMol can use PG to adjust atom positions v_t for minimizing steric clashes between generated molecules and protein pockets. **d**, Shape-conditioned molecule prediction module SMP. DIFF uses SMP to predict the atom positions and features $(\tilde{x}_{0,t}, \tilde{v}_{0,t})$ given the noisy data (x_t, v_t) and $\mathbf{H}^s$. SMP is a multilayer gNN comprising L layers. In the l th layer, SMP leverages a SARL module, a BTRL module and a GVP to jointly learn effective atom representations for the prediction. **e**, Shape guidance SG. During the generation, DiffSMol can use SG to explicitly push predicted atoms to the shapes of condition molecules.

for new protein targets. Recently, generative artificial intelligence (genAI) methods, such as variational autoencoders⁷, diffusion⁸ and ChatGPT⁹, have emerged as groundbreaking computational tools for many applications¹⁰, including drug design^{11–13}. Instead of searching for drug candidates, genAI methods could directly generate molecules satisfying prescribed properties (for example, lipophilicity and drug likeness) through learning the underlying chemical knowledge carried by vast molecule datasets and making autonomous decisions in constructing new molecules^{11,12} (for example, molecular graphs and three-dimensional (3D) structures). The powerful generative capabilities of genAI demonstrate great promise in fundamentally transforming the traditional drug development process into a more focused, accurate, swift and sustainable alternative.

Here we introduce DiffSMol, a genAI method to generate small molecules in three dimensions that effectively bind to a protein target and have realistic structures (for example, correct bond angles and bond lengths). For succinctness, unless otherwise specified, all molecules referred to in the following sections are small molecules. Motivated by LBDD, DiffSMol generates binding molecules based on the shapes of known ligands, following the principle that molecules with similar shapes tend to have similar binding activities^{4,14}. DiffSMol encapsulates the geometric details of ligand shapes within pretrained, expressive shape embeddings and generates binding molecules, including their atom types and atom positions through diffusion⁸. During the iterative diffusion process, DiffSMol leverages a molecule graph representation learning approach and integrates ligand shape

embeddings in generating and refining the atom types and atom positions and, thus, a de novo molecule and its 3D structures. To better resemble the known ligand shapes, DiffSMol further modifies the generated 3D structures iteratively under the guidance of the ligand shapes. Inspired by SBDD, in addition to ligands, DiffSMol can also leverage the geometric information of protein binding pockets and tailor the generated molecules towards optimal binding affinities under the guidance of binding pockets. Fig. 1 shows the overview architecture of DiffSMol. Table 1 summarizes the notations used in this article.

In summary, DiffSMol has the following advantages:

- DiffSMol generates molecules that closely retain the shapes of known ligands but have entirely different molecule structures, by utilizing a two-stage framework. It first encodes ligand shapes into expressive shape embeddings and then generates small molecules with similar shapes to ligands based on these embeddings.
- DiffSMol pretrains expressive shape embeddings that capture the geometric details of ligand shapes. The shape embeddings are calculated by an equivariant shape encoder and optimized by recovering the signed distances of randomly sampled points in 3D space to the molecule surface based on the shape embeddings.
- DiffSMol fully integrates ligand shape embeddings in generating and refining the atom types and atom coordinates during the diffusion process via a molecule graph representation learning approach.

Table 1 | Notations

Notation	Meaning
M	a molecule
a_i	the i th atom in M
$\mathbf{x}_i$	the position of a_i in 3D space
$\mathbf{v}_i$	the feature vector of a_i
d_{ij}	the distance between a_i and a_j
$\mathbf{b}_{ij}$	the one-hot feature vector indicating the bond type between a_i and a_j
s	the 3D surface shape of M
$\mathcal{P}$	the point cloud for s
z_i	the i th point in $\mathcal{P}$
$\mathbf{H}_s$	the latent embedding of $\mathcal{P}$
o	the signed distance of a point randomly sampled in 3D space to the molecule surface

- DiffSMol introduces shape guidance to iteratively modify the generated 3D structures of small molecules to better resemble the shape of the ligands. DiffSMol also develops protein guidance to utilize the geometric information of protein binding pockets and tailor the generated small molecules for strong binding.
- DiffSMol outperforms state-of-the-art shape-conditioned molecule generation (SMG) methods by 153.6% improvement in the success rate of generating desirable molecules with highly similar shapes to ligands. Incorporating shape guidance further boosts the improvement to 448.2%.
- DiffSMol with pocket guidance achieves comparable or even superior performance compared with state-of-the-art pocket-conditioned molecule generation (PMG) methods in generating realistic binding molecules, with similar binding affinities and substantially higher drug-likeness values. Compared with existing PMG methods, DiffSMol is at least ten times faster in the average time cost of generating a single molecule.
- Case studies with extensive in silico analyses for two important drug targets demonstrate that DiffSMol effectively generates molecules with high binding affinities, favourable drug-like properties and desirable ADMET (absorption, distribution, metabolism, excretion and toxicity) profiles, suggesting the potential of DiffSMol in advancing drug development.

Related work

A variety of deep generative models have been developed to generate molecules using various molecule representations, including generating SMILES string representations¹⁵, or two-dimensional (2D) molecular graph representations^{11,16}. However, these representations fall short in capturing the 3D structures of molecules, which are critical for understanding their biological activities and certain properties. Recent efforts have been dedicated to the generation of 3D molecules. For example, Hoogetboom et al.¹⁷ developed an equivariant diffusion model in which an equivariant network is used to jointly predict both the positions and features of all atoms. In 3D molecule generation, two types of method have been developed. Motivated by LBDD, the first type of method, referred to as SMG methods, generates molecules with similar shapes to condition molecules (for example, ligands). The second type of method, referred to as PMG methods, is motivated by SBDD and generates binding molecules to a target protein pocket. Due to space constraints, only SMG methods are discussed below and other related works are presented in Supplementary Section 1. For example, PMG methods are discussed in Supplementary Section 1.1, and scaffold hopping methods and their comparison with SMG methods in inputs, objectives and methodologies are discussed in Supplementary Section 1.2.

SMG methods

Previous SMG methods^{18,19} generally leverage shapes as conditions and use generative models such as variational autoencoders⁷ to generate potentially binding molecules. Among SMG methods, Adams and Coley¹⁹ developed SQUID, which consists of a fragment-based generative model and a rotatable-bond scoring model. The former generates molecules using variational autoencoders and sequentially decodes fragments based on the shapes of condition molecules (for example, ligands), while the latter adjusts the angles of rotatable bonds between fragments to adapt to the condition shapes. Long et al.¹⁸ developed an encoder–decoder framework, referred to as Shape2Mol, that first encodes 3D shapes of molecules into latent embeddings and then generates fragments sequentially on the basis of these embeddings to build molecules. In our preliminary work²⁰, we also demonstrated the potential of diffusion models for generating binding molecules conditioned on shapes. In this Article, we have substantially extended our previous research in methodology, experiments and analyses. By improving the shape-conditioned molecule prediction module (see ‘Shape-conditioned molecule prediction’ section in Methods), we have substantially enhanced the performance of DiffSMol in SMG. In addition, the integration of protein pocket guidance (see ‘DiffSMol with protein pocket guidance’ section in Methods) extends the applicability of DiffSMol beyond SMG to PMG, where it also outperforms existing PMG methods. Case studies with extensive analyses further highlight the potential of DiffSMol to generate molecules with high binding affinities and desirable properties. Detailed discussions on the comparison between DiffSMol and the above SMG methods are available in Supplementary Section 1.3.

Experimental results

We evaluate the effectiveness of DiffSMol in both SMG and PMG. For SMG, following the literature¹⁹, we evaluate whether DiffSMol could generate realistic 3D molecules that have shapes similar to condition molecules; for PMG, following the literature^{21–23}, we assess, given target protein pockets, whether DiffSMol could generate molecules with high binding affinities and realistic structures. In particular, for SMG, we evaluate DiffSMol and its variant with shape guidance (detailed in ‘DiffSMol with shape guidance’ section in Methods), referred to as DiffSMol+s, against the state-of-the-art SMG baselines in terms of shape similarity, diversity and realism of generated molecules. For PMG, we compare both DiffSMol and DiffSMol+s with pocket guidance (detailed in ‘DiffSMol with pocket guidance’ section in Methods), referred to as DiffSMol+p and DiffSMol+s+p, with state-of-the-art PMG baselines to investigate if DiffSMol can effectively generate realistic molecules binding towards protein targets. Note that, different from PMG baselines trained on sparse protein–ligand complex data, DiffSMol is capable of leveraging large-scale molecule data for better generation. Details about the SMG and PMG baselines, the data used in our experiments, the experimental set-ups and the evaluation metrics are available in Supplementary Section 2. Details about hyperparameters used in DiffSMol are available in Supplementary Section 3.

Overall comparison on generating desirable molecules in SMG

We evaluate DiffSMol, DiffSMol+s and state-of-the-art SMG baselines in generating desirable molecules. Following Bostroem et al.¹⁴, we define desirable molecules as those satisfying the 2D graph similarity constraint δ_g and with 3D shape similarities larger than 0.8 (detailed in Supplementary Section 2.4). These molecules have highly similar shapes to the condition molecules (for example, ligands) and, thus, could also have desirable binding activities⁴. For each method, we calculate the possibility of generating desirable molecules (that is, #d%), the diversity and the novelty among these molecules (that is, Div_d and #n%, respectively) under different graph similarity constraints (that is, $\delta_g = 0.3, 0.5, 0.7$ and 1.0). As shown in Table 2, DiffSMol and DiffSMol+s consistently outperform all the baselines in terms of all

Table 2 | Comparison of desirable molecules for SMG

Method	$\delta_g = 0.3$			$\delta_g = 0.5$			$\delta_g = 0.7$			$\delta_g = 1.0$		
	#d%	Div _d	#n%	#d%	Div _d	#n%	#d%	Div _d	#n%	#d%	Div _d	#n%
VS	10.6	0.736	0.0	12.2	0.734	0.0	12.3	0.734	0.0	12.3	0.734	0.0
SQUID($\lambda = 0.3$)	8.3	0.669	96.6	21.8	0.649	96.2	27.5	0.633	95.7	31.3	0.617	92.6
SQUID($\lambda = 1.0$)	11.2	0.728	96.9	14.4	0.721	96.7	14.6	0.720	96.6	14.7	0.720	96.6
DiffSMol	<u>28.4</u>	0.762	<u>99.8</u>	<u>32.3</u>	0.751	<u>99.8</u>	<u>32.4</u>	0.751	<u>99.8</u>	<u>32.4</u>	0.751	<u>99.8</u>
DiffSMol+s	61.4	<u>0.760</u>	99.9	70.9	<u>0.748</u>	99.9	71.0	<u>0.748</u>	99.9	71.0	<u>0.748</u>	99.9

δ_g , the 2D graph similarity constraint; #d%, the percentage of generated molecules that are desirable, satisfying δ_g and exhibiting high 3D shape similarity larger than 0.8; Div_d, the diversity among the desirable molecules; #n%, the percentage of desirable molecules that cannot be found in the MOSES dataset. The best values are in bold, and the second-best values are underlined.

the metrics. For example, when $\delta_g = 0.3$, at #d%, DiffSMol+s (61.4%) demonstrates a statistically significant ($P < 0.001$) improvement of 448.2% compared with the best baseline SQUID ($\lambda = 1.0$) (11.2%). In terms of Div_d, DiffSMol (0.762) also significantly outperforms the best baseline VS (0.736), a virtual screening method, by 2.3% ($P < 0.001$). At #n%, both DiffSMol and DiffSMol+s ensure that nearly all the generated desirable molecules are highly different from the training molecules (99.8% for DiffSMol and 99.9% for DiffSMol+s), substantially outperforming the best baseline SQUID with $\lambda = 1.0$ (96.9%) by 3.1% and 3.0%, respectively. When $\delta_g = 0.5, 0.7$ or 1.0 , a similar trend is observed. At Div_d and #n%, DiffSMol and DiffSMol+s also demonstrate top performance among all the methods. The superior performance of DiffSMol and DiffSMol+s in #d%, Div_d and #n%, particularly at small δ_g , indicates their strong capacity in generating de novo molecules that have desirable shapes and distinct graph structures compared with the condition molecules, thereby facilitating the process of drug development.

It is worth noting that, as shown in Table 2, DiffSMol and DiffSMol+s consistently outperform SQUID with $\lambda = 0.3$ and 1.0 in terms of all the metrics. A key distinction between DiffSMol and SQUID is that SQUID generates molecules by sequentially predicting fragments. However, during fragment prediction, their poses are not fully considered, leading to suboptimal prediction accuracy and limited generation performance. Meanwhile, by directly arranging atoms, DiffSMol and DiffSMol+s explicitly consider the 3D atom positions when generating molecules and, thus, achieve remarkable improvement over SQUID as shown in Table 2.

Different from DiffSMol and SQUID, which directly generate desirable molecules, VS screens over randomly sampled training molecules to identify molecules of interest. However, it cannot ensure optimal precision design, resulting in the suboptimal performance of VS at #d%. In addition, due to the reliance on existing molecules, VS cannot discover molecules that are not in the library. By contrast, DiffSMol can effectively generate de novo molecules with desirable shapes, making it a promising tool for discovering drug candidates.

Comparing DiffSMol+s and DiffSMol, Table 2 shows that incorporating shape guidance into DiffSMol substantially boosts its effectiveness in generating desirable molecules. For example, when $\delta_g = 0.3$, at #d%, DiffSMol+s (61.4%) significantly outperforms DiffSMol (28.4%) by 116.2% with $P < 0.001$. When $\delta_g = 0.5, 0.7$ and 1.0 , DiffSMol+s also achieves a considerable improvement of 119.5%, 119.1% and 119.1%, respectively, compared with DiffSMol, while retaining very similar performance in terms of the diversity and novelty. These results signify that shape guidance effectively improves the ability of DiffSMol in generating molecules that have similar shapes to condition molecules without degrading the novelty and diversity among generated molecules.

Supplementary Section 4 presents further results on the comparison of shape and graph similarity (Supplementary Section 4.1), the comparison of molecule validity and novelty (Supplementary Section 4.2), the analyses of the distributions of shape and graph similarities (Supplementary Section 4.3), case studies on generated molecules

from DiffSMol and baselines (Supplementary Section 4.4) and case studies on 200 generated molecules (Supplementary Section 4.5). Additional case studies, including analyses of low-quality molecules (for example, unrealistic structures or low shape similarities), are provided in Supplementary Section 5.

Overall comparison for PMG

From this section, we shift our focus from evaluating against SMG baselines to PMG baselines (see Supplementary Section 1.1 for details). We evaluate the effectiveness of DiffSMol against state-of-the-art PMG baselines (see Supplementary Section 2.1 for details) in generating molecules binding towards specific protein pockets. All the baselines require protein–ligand complex data for training and generate molecules by explicitly modelling their interactions with protein binding pockets. Different from these baselines, DiffSMol does not require complex data and consumes molecules for training. By contrast, there exist several high-quality and large-scale molecule databases^{24,25}. By consuming molecule data for training, DiffSMol could fully leverage the rich data for better generation. DiffSMol further enhances the binding molecule generation by incorporating pocket guidance as detailed in ‘DiffSMol with pocket guidance’ section in Methods.

We utilize two variants of DiffSMol for evaluation: DiffSMol with pocket guidance (DiffSMol+p) and DiffSMol with both pocket and shape guidance (DiffSMol+s+p). In this section, we evaluate DiffSMol+p, DiffSMol+s+p and PMG baselines in both effectiveness and efficiency. Following previous work^{23,26}, in terms of the effectiveness, we evaluate the binding affinity, drug likeness and diversity of molecules generated from DiffSMol+p, DiffSMol+s+p and all PMG baselines. Please refer to Supplementary Section 2.4 for a detailed description for the evaluation metrics. Regarding efficiency, we report the inference time of all methods used to generate molecules.

We notice that DiffSMol+p and DiffSMol+s+p show remarkable efficiency over baselines by using the computationally lightweight pocket guidance PG and the pretrained shape embedding module SE. The pocket guidance requires only minimal adjustments to atom positions. The pretrained SE allows DiffSMol to precalculate and compress shape conditions into latent embeddings once and, thus, avoid repeated recalculations for each generated molecule. Consequently, DiffSMol achieves faster generation than diffusion-based PMG methods that jointly learn pocket conditions and molecule structures at each generation step. The superior efficiency enables DiffSMol+p and DiffSMol+s+p to generate more than 10× the molecules than baselines in the same time duration. Therefore, following Long et al.¹⁸, for baselines, we apply them to generate 100 molecules for each test protein target. For DiffSMol, we generate 1,000 molecules and select the top 100 molecules for comparison based on their Vina S, quantitative estimate of drug-likeness (QED) and synthetic accessibility (SA) scores. We report the performance of all methods in Table 3. This comparison aims to validate that, compared with baselines, DiffSMol could generate small molecules with higher binding affinities while using less computing as indicated by the inference time in Table 3. For completeness, we

Table 3 | Overall comparison on PMG

Method	Vina S↓		Vina M↓		Vina D↓		HA%↑		QED↑		SA↑		Lipinski↑		Div↑		Time (s)↓	
	Avg.	Med.	Avg.	Med.	Avg.	Med.	Avg.	Med.	Avg.	Med.	Avg.	Med.	Avg.	Med.	Avg.	Med.	Total	Per
Reference	-5.32	-5.66	-5.78	-5.76	-6.63	-6.67	-	-	0.53	0.49	0.77	0.77	4.69	5.00	-	-	-	-
AR	-5.06	-4.99	-5.59	-5.29	-6.16	-6.05	37.69	31.00	0.50	0.49	0.66	0.65	4.81	5.00	0.70	0.70	7,789	77.89
Pocket2Mol	-4.50	-4.21	-5.70	-5.27	-6.43	-6.25	48.00	51.00	0.58	0.58	0.77	0.78	4.95	5.00	0.69	0.71	2,150	21.50
DiffSBDD	8.81	-3.46	0.82	-5.11	-1.67	-7.14	54.19	58.00	0.48	0.48	0.56	0.56	4.64	5.00	<u>0.76</u>	<u>0.75</u>	679	6.79
TargetDiff	-4.88	<u>-5.82</u>	-6.20	<u>-6.36</u>	-7.37	<u>-7.51</u>	57.57	58.27	0.50	0.51	0.60	0.59	4.62	5.00	0.72	0.71	1,252	12.52
DecompDiff	-4.58	-4.77	-5.47	-5.51	-6.43	-6.56	47.76	48.66	0.56	0.56	0.70	0.69	4.80	5.00	0.72	0.72	1,859	18.59
PMDM	4.71	-3.07	-1.99	-5.57	-7.12	-7.50	53.35	55.28	0.55	0.56	0.62	0.61	4.77	5.00	0.78	0.77	572	5.72
DiffSMol+p	<u>-5.53</u>	-5.64	<u>-6.37</u>	-6.33	<u>-7.19</u>	-7.52	<u>78.75</u>	94.00	0.77	0.80	<u>0.76</u>	<u>0.76</u>	4.99	5.00	0.63	0.66	462	0.46
DiffSMol+s+p	-5.81	-5.96	-6.50	-6.58	-7.16	<u>-7.51</u>	79.92	<u>93.00</u>	<u>0.76</u>	<u>0.79</u>	0.75	0.74	4.99	5.00	0.64	0.66	561	0.56

Vina S, the binding affinities between the initially generated poses of molecules and the protein pockets; Vina M, the binding affinities between the poses after local structure minimization and the protein pockets; Vina D, the binding affinities between the poses determined by AutoDock Vina²⁴ and the protein pockets; HA, the percentage of generated molecules with Vina D higher than those of condition molecules; QED, the drug-likeness score; SA, the synthesizability score; Lipinski, the Lipinski's rule of five; Div, the diversity among generated molecules; Time (s), the average total time cost in seconds of generating molecules per protein target and the average time cost per molecule. Avg., average value of the metrics calculated over all generated molecules; Med., median value of the metrics calculated over all generated molecules. For baseline methods, 'Total' represents the time to generate 100 molecules, while for DiffSMol+p and DiffSMol+s+p, it represents the time to generate 1,000 molecules. The best values are in bold, and the second-best values are underlined.

present the results of DiffSMol+p and DiffSMol+s+p when generating 100 molecules in Supplementary Section 6, with detailed discussions in Supplementary Section 6.1. This comparison focuses purely on the binding affinities of small molecules generated by different methods and does not take into account the computing required (for example, inference time).

As shown in Table 3, DiffSMol+p and DiffSMol+s+p achieve the second-best and best performance in terms of the binding affinities of generated molecules. In particular, they demonstrate the second-best ($-5.53 \text{ kcal mol}^{-1}$) and best ($-5.81 \text{ kcal mol}^{-1}$) average Vina S, with statistically significant improvement of 9.3% and 14.8% over the best baseline AR ($-5.06 \text{ kcal mol}^{-1}$) ($P < 0.001$ for both comparison). For average Vina scores obtained from locally minimized poses (that is, Vina M) and poses optimized by global search (that is, Vina D), both DiffSMol+p and DiffSMol+s+p individually show no statistically significant difference from the best baseline TargetDiff for both Vina M and Vina D ($P > 0.1$ for all comparisons). These results demonstrate that, even without explicitly training on protein–ligand complexes, DiffSMol+p and DiffSMol+s+p could still generate molecules with comparable or superior binding affinities towards protein targets in terms of Vina S, Vina M and Vina D, compared with state-of-the-art PMG baselines. Note that we consider Vina M and Vina D in evaluation, as it is a common practice in drug development to find more favourable binding poses of candidates through pose optimization²⁷. We observe that the average Vina S and Vina M of DiffSBDD and PMDM are notably worse compared with the other methods. This could be due to these models struggling to fully incorporate comprehensive protein structural information into their generative process, resulting in generated molecules that often have unfavourable steric clashes and low binding affinities to protein targets.

When comparing DiffSMol+p and DiffSMol+s+p, Table 3 shows that, overall, DiffSMol+s+p with shape guidance can generate molecules with higher binding affinities compared with DiffSMol+p. To be specific, for Vina S and Vina M, DiffSMol+s+p significantly outperforms DiffSMol+p by 5.1% and 2.0% ($P < 0.001$), respectively. At Vina D, the performance of DiffSMol+s+p and DiffSMol+p is highly comparable. These results indicate that, even with pocket guidance, including additional shape guidance could further enhance the generation of binding molecules. Table 3 also shows that DiffSMol+p and DiffSMol+s+p achieve the state-of-the-art performance in generating molecules of higher binding affinities than condition molecules (HA), favourable drug likeness (QED and Lipinski), synthesizability (SA) and diversity (Div). Detailed discussions on comparison between DiffSMol and baselines

regarding these metrics are available in Supplementary Section 6.2. We also compare the quality of generated molecules, including stability, 3D structures and 2D structures, from DiffSMol+p, DiffSMol+s+p and baselines in Supplementary Section 6.3. Case studies on molecules with low binding affinities to protein targets generated by DiffSMol are available in Supplementary Section 5.3.

Case studies for targets

DiffSMol can generate binding molecules that serve as promising drug candidates. To demonstrate this ability, we highlight three molecules generated by DiffSMol+s+p for two crucial drug targets, cyclin-dependent kinase 6 (CDK6) and neprilysin (NEP). CDK6 plays a critical role in cell proliferation by regulating cell cycle progression. Inhibiting CDK6 can disrupt the abnormal cell cycles of cancer cells, making it a valuable therapeutic target for cancers²⁸. NEP can help prevent amyloid plaque formation associated with Alzheimer's disease, making it an important target for therapies to potentially slow the disease's progression²⁹. We use an existing protein–ligand complex for CDK6 (PDBID:4AUA) and an existing protein–ligand complex for NEP (PDBID:1RIH), respectively, from Protein Data Bank (PDB)³⁰ and generate 1,000 molecules for each of the targets. Both complexes are included in our test set for PMG. For each target, we prioritize the best molecule based on their Vina S, QED³¹, SA³², toxicity scores calculated by ICM³³ and absorption, distribution, metabolism, excretion and toxicity (ADMET) metrics calculated by admetSAR 2.0 (ref. 34). We select two top drug candidates for CDK6 and one top drug candidate for NEP. Note that the 3D structures of these molecules and their binding poses are generated by DiffSMol+s+p without any postprocessing such as energy minimization or docking. All the generated 3D structures are validated to be realistic by their close match, with a root mean square deviation of less than 2 Å, to minimized structures from the Cartesian Merck molecular force field (MMFF) minimization algorithm³⁵. Due to space constraints, we present only two generated molecules for CDK6 in this Article and discuss additional case study results in Supplementary Section 7 (for the generated molecule for NEP, see Supplementary Section 7.1).

Generated molecules for CDK6. Figure 2 shows two generated molecules for CDK6. In particular, Fig. 2a shows a generated molecule, referred to as NL-001, for CDK6 and its structures and binding interactions with the CDK6 binding pocket. As shown in Fig. 2a, molecule NL-001 fits well within the CDK binding pocket and forms hydrophobic interactions with the surrounding residues, such as T107,

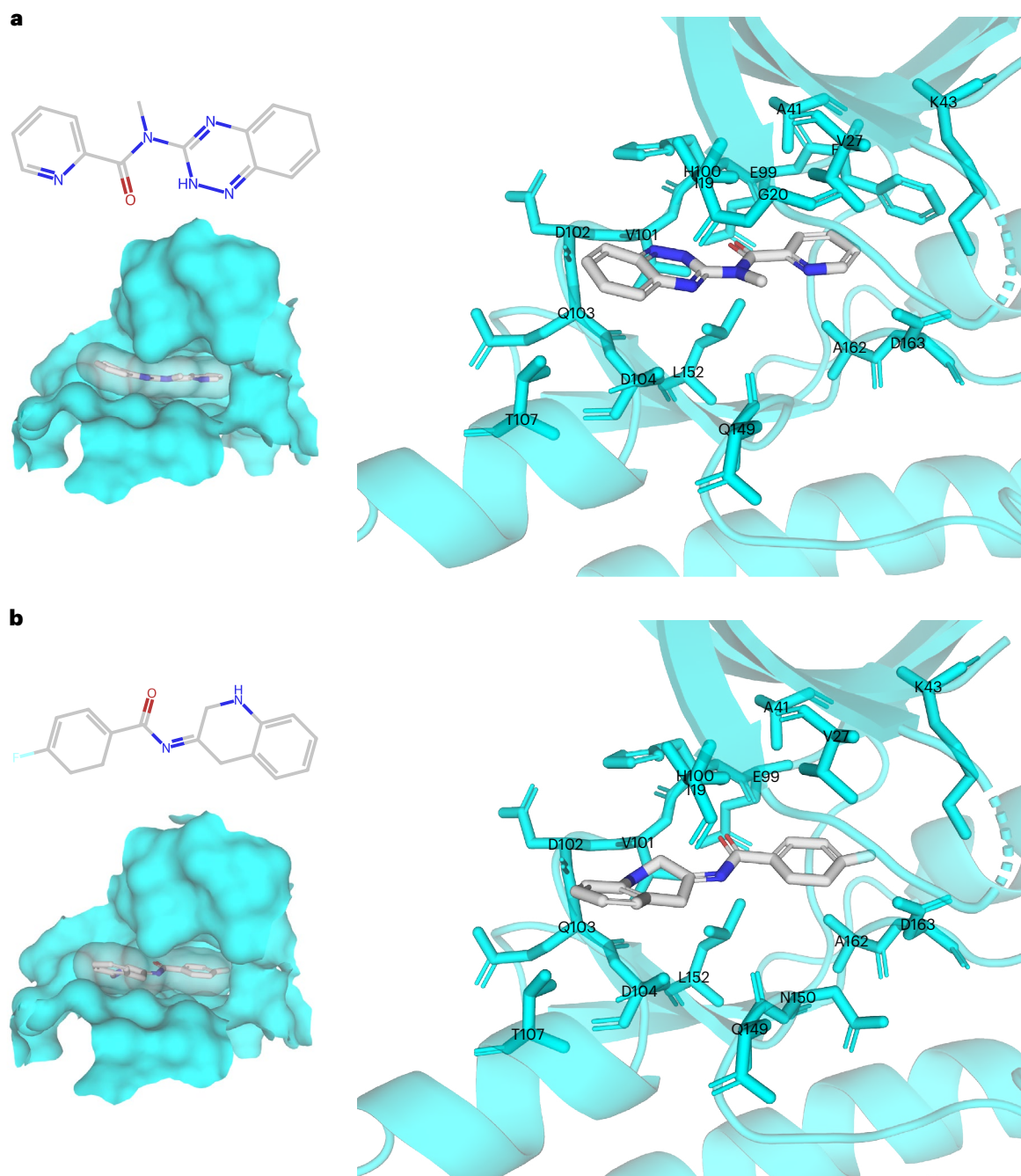


Fig. 2 | Generated drug candidates for CDK6. **a**, Generated drug candidate NL-001 for CDK6, each with a 2D molecular graph, a surface representation of NL-001 and CDK6 interaction and a cartoon representation; **b**, Generated drug candidate NL-002 for CDK6, each with a 2D molecular graph, a surface representation of NL-001 and CDK6 interaction and a cartoon representation.

D104, L152 and A162. This effective binding results in a better Vina S of $-6.817 \text{ kcal mol}^{-1}$ for the generated molecule, compared with the Vina S ($0.736 \text{ kcal mol}^{-1}$) of the 4AU ligand in the complex 4UAU. Local minimization and docking refinement can further improve the Vina score of this molecule to $-7.251 \text{ kcal mol}^{-1}$ (Vina M) and $-8.319 \text{ kcal mol}^{-1}$ (Vina D), respectively, outperforming the 4AU ligand ($-5.939 \text{ kcal mol}^{-1}$ for Vina M and $-7.592 \text{ kcal mol}^{-1}$ for Vina D).

In addition to the binding activity, the molecule in Fig. 2a also demonstrates favourable properties that are important for drug development, including drug likeness, synthesizability, toxicity and ADMET profiles. This molecule meets the Lipinski rule of five criteria³⁶, with a QED score of 0.834, higher than that of 4AU ligand (0.773). Its SA score of 0.720 suggests favourable synthesizability. This molecule also has

a low toxicity score (0.236). To fully evaluate its potential as a drug candidate, we compare its ADMET profile with those of three Food and Drug Administration-approved CDK6 inhibitors, abemaciclib³⁷, palbociclib³⁸ and ribociclib³⁹. The results show that our molecule has comparable or even better ADMET properties in metrics crucial for cancer drug development, compared with those approved drugs. For example, same as the approved drugs, our molecule is predicted to be negative for carcinogenicity⁴⁰ and nephrotoxicity⁴¹. Notably, our molecule has a higher score than all the approved drugs in plasma protein binding, indicating its capacity to be distributed throughout the body and reach the target site. Details about the properties of the generated molecule NL-001 for CDK6 are available in Supplementary Sections 7.2 and 7.3.

Figure 2b shows another promising drug candidate for CDK6 generated by DiffSMol, referred to as NL-002. It has very similar properties to NL-001, with a Vina S score of $-6.970 \text{ kcal mol}^{-1}$, a Vina M score of $-7.605 \text{ kcal mol}^{-1}$ and a Vina D score of $-8.986 \text{ kcal mol}^{-1}$, showing its strong binding affinity to CDK6. Notably, NL-002 has a low toxicity score (0.000) and does not have any known toxicity-inducing functional groups detected³³. NL-002 also has a very similar ADMET profile as NL-001, suggesting it could be another strong drug candidate for CDK6. Details about the properties of the generated molecule NL-002 for CDK6 are available in Supplementary Sections 7.2 and 7.3.

Discussion and conclusion

In this section, we discuss the current limitations of DiffSMol, identify promising future research directions and summarize the findings of this Article. We highlight opportunities to integrate protein structural information in binding molecule generation. Further discussions on comprehensive evaluation, interpretable molecule generation, multiobjective molecule optimization and in vitro validation for in silico generated molecules are available in Supplementary Sections 8–11.

Integrating protein targets for binding molecule generation

For PMG, our experiments show that DiffSMol with pocket guidance (DiffSMol+p and DiffSMol+s+p) can effectively generate molecules with high binding affinities towards protein targets. As detailed in ‘DiffSMol with protein pocket guidance’ section in the Methods, this pocket guidance enables DiffSMol+p and DiffSMol+s+p to consider geometric information of protein binding pockets when generating binding molecules. In addition to geometric information, we acknowledge that incorporating other information about protein pockets could further enable molecules generated in high quality. For example, physicochemical properties of amino acid sequences within binding pockets, such as polarity, electrostatics and hydrophobicity, can affect the strength of interactions between proteins and molecules⁴². Incorporating information about the dynamic nature of binding pockets, such as potential conformational changes or the presence of flexible loops, could further lead to molecules with improved binding affinities. Therefore, a generative model considering these properties could produce molecules that better conform to what is expected on the basis of pharmaceutical chemistry, shortening their pathways to be induced into downstream tasks of drug development. Identifying and integrating essential properties of protein binding pockets into molecule generation could be an interesting yet challenging future research direction, although out of the scope of this study.

Conclusions

DiffSMol generates binding molecules with realistic 3D structures based on the shapes of known ligands. It utilizes pretrained shape embeddings and a customized diffusion model for binding molecule generation. To better resemble the known ligand shapes, DiffSMol also modifies the generated 3D molecules iteratively under the guidance of the ligand shapes. In addition, it can leverage geometric information of protein binding pockets and tailor the generated molecules towards optimal binding affinities. Experimental results demonstrate that DiffSMol outperforms SMG methods in generating molecules with highly similar shapes to known ligands, while incorporating shape guidance further boosts this performance. Compared with PMG methods, DiffSMol with pocket guidance also achieves exceptional performance in generating molecules with high binding affinities. The case studies involving two critical drug targets show that DiffSMol can generate binding molecules with desirable drug properties. However, DiffSMol still has limitations. In addition to the limitations and corresponding future research directions that have been discussed above, one limitation with DiffSMol is that the binding poses of generated molecules are typically constrained by those of known ligands. This limitation can confine the ability of DiffSMol to explore unseen binding poses. Thus, a future research

direction is on how to mitigate this limitation by inferring diverse ligand shapes from protein pockets. Another possible research direction is to extend the applications of DiffSMol to additional tasks, such as scaffold hopping and molecule optimization, to fully explore the potential of DiffSMol and its applicability in drug development.

Methods

DiffSMol aims to generate binding molecules based on the shapes of known ligands, following the principle that molecules with similar shapes tend to have similar binding activities. Towards this end, DiffSMol consists of two modules: (1) a pretrained equivariant shape embedding module SE that learns expressive latent embeddings for the shapes of condition molecules (for example, ligands) and (2) an equivariant molecule diffusion model DIFF that explicitly considers shape embeddings from SE to generate 3D molecules with similar shapes to condition molecules. In particular, given a condition molecule, SE represents its shape as a point cloud with points sampled over its molecular surface. SE learns to map this point cloud into a latent embedding $\mathbf{H}^s$ using an encoder–decoder framework (more details in ‘Equivariant condition shape representation pretraining’ section). Conditioned on the shape embedding $\mathbf{H}^s$, DIFF learns to generate molecules with desired shapes and realistic topologies in an equivariant way. In particular, DIFF utilizes equivariant graph neural networks (GNNs) to learn shape-aware atom embeddings and generate molecules tailored to the shape condition. DIFF also leverages bond types as a training signal to fully capture the inherent topologies of molecules. During inference, DIFF utilizes shape guidance to further direct the generated molecules towards the shape condition. Besides shape guidance, when the structure of the protein binding pocket is available, DIFF uses pocket guidance to adjust the atom positions of generated molecules for optimal binding affinities with the binding pocket. This design enables the applicability of DiffSMol for PMG. Figure 1 shows the overall architecture of DiffSMol. All the algorithms are presented in Supplementary Section 12. We introduce the concepts of equivariance and invariance in detail in Supplementary Section 13.

In the following sections, we first introduce the key notations and the definitions of equivariance and invariance in ‘Representations and notations’ section. We then introduce the equivariant shape embedding module SE in ‘Equivariant condition shape representation pretraining’ section. After that, we discuss the shape-conditioned molecule diffusion model DIFF in ‘Diffusion-based molecule generation’ section. We describe the shape-conditioned molecule prediction module SMP used in DIFF in ‘Shape-conditioned molecule prediction’ section. Finally, we describe the molecule generation process and how the shape guidance and pocket guidance are used during inference in ‘Guidance-induced inference’ section.

Representations and notations

This section is largely based on our preliminary work²⁰. We represent a molecule M as a set of atoms $M = \{a_1, a_2, \dots, a_{|M|} | a_i = (\mathbf{x}_i, \mathbf{v}_i)\}$, where $|M|$ is the number of atoms in M ; a_i is the i th atom in M ; $\mathbf{x}_i \in \mathbb{R}^3$ represents the position of a_i in 3D space; and $\mathbf{v}_i \in \mathbb{R}^k$ is a_i ’s one-hot atom feature vector indicating the atom type and its aromaticity. We represent the Euclidean distance between each pair of atoms a_i and a_j as $d_{ij} \in \mathbb{R}$, and the type of the bond in between as a one-hot vector $\mathbf{b}_{ij} \in \mathbb{R}^4$, in which the four dimensions of $\mathbf{b}_{ij}$ represent the absence of a bond, a single bond, a double bond and an aromatic bond, respectively. Following Guan et al.²³, bonds between atoms can be uniquely determined by the atom types and the atomic distances among atoms. We represent the 3D surface shape s of a molecule M as a point cloud constructed by sampling points over the molecular surface. Details about the construction of point clouds from the surface of molecules are available in Supplementary Section 14. We denote the point cloud as $\mathcal{P} = \{z_1, z_2, \dots, z_{|\mathcal{P}|} | z_j = (\mathbf{z}_j)\}$, where $|\mathcal{P}|$ is the number of points in $\mathcal{P}$; z_j is the j th point; and $\mathbf{z}_j \in \mathbb{R}^3$ represents the position of z_j in 3D space. We

denote the latent embedding of $\mathcal{P}$ as $\mathbf{H}^s \in \mathbb{R}^{d_p \times 3}$, where d_p is the dimension of the latent embedding. We represent the distance of a point randomly sampled in 3D space to the molecule surface as o , referred to as a signed distance, with a positive (negative) sign indicating the point is inside (outside) the surface. Table 1 summarizes the notations used in this Article.

Equivariant condition shape representation pretraining

This section is largely based on our preliminary work²⁰. DiffSMol pretrains a shape embedding module SE to generate surface shape embeddings $\mathbf{H}^s$ of condition molecules. SE uses an encoder SE-enc to map $\mathcal{P}$ to the equivariant latent embedding $\mathbf{H}^s$. SE uses a decoder SE-dec to optimize $\mathbf{H}^s$ by recovering the signed distances⁴³ of randomly sampled points in 3D space to the molecule surface using $\mathbf{H}^s$. DiffSMol uses $\mathbf{H}^s$ to guide the diffusion process as will be detailed later (see ‘Diffusion-based molecule generation’ section). We present SE in detail in the following sections. In particular, we present the encoder SE-enc in ‘Shape encoder’ section; the decoder SE-dec in ‘Shape decoder’ section; and the optimization of SE in ‘SE pretraining’ section. Figure 1a shows the architecture of SE.

Shape encoder. SE-enc learns shape embeddings $\mathbf{H}^s$ from the 3D surface shape $\mathcal{P}$ of molecules in an equivariant way. To ensure translation equivariance, SE-enc shifts the centre of each $\mathcal{P}$ to zero to eliminate all translations. To ensure rotation equivariance, SE-enc leverages Deng et al.’s vector neuron (VN)-based encoder⁴⁴, VN-DGCNN, to learn equivariant shape embeddings $\mathbf{H}^s$ as follows:

$$\{\mathbf{H}_1^p, \mathbf{H}_2^p, \dots, \mathbf{H}_{|\mathcal{P}|}^p\} = \text{VN-DGCNN}(\{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{|\mathcal{P}|}\}),$$

$$\mathbf{H}^s = \sum_j \mathbf{H}_j^p / |\mathcal{P}|, \quad (1)$$

where VN-DGCNN($\cdot$) is a VN-DGCNN⁴⁵ to generate equivariant embedding $\mathbf{H}_j^p \in \mathbb{R}^{3 \times d_p}$ for each point $\mathbf{z}_j$ in $\mathcal{P}$; and $\mathbf{H}^s \in \mathbb{R}^{3 \times d_p}$ is the embedding of $\mathcal{P}$ generated via a mean pooling over the embeddings of all the points. VN-DGCNN($\cdot$) guarantees the rotation equivariance by learning embedding matrices $\mathbf{H}_j^p \in \mathbb{R}^{3 \times d_p}$ for points using only equivariant operations as detailed in Deng et al.⁴⁴.

Shape decoder. To optimize $\mathbf{H}^s$, following Deng et al.⁴⁴, SE learns a decoder SE-dec to predict the signed distance of a query point $\mathbf{z}_q$ randomly sampled from 3D space to surface shape $\mathcal{S}$ using multilayer perceptrons (MLPs) as follows:

$$\hat{o}_q = \text{MLP}([\langle \mathbf{z}_q, \mathbf{H}^s \rangle, \|\mathbf{z}_q\|^2, \text{VN-In}(\mathbf{H}^s)]), \quad (2)$$

where $\hat{o}_q$ is the predicted signed distance of $\mathbf{z}_q$, with positive and negative values indicating that $\mathbf{z}_q$ is inside or outside the surface shape $\mathcal{S}$, respectively; $[\cdot, \cdot]$ represents the concatenation operation; $\langle \cdot, \cdot \rangle$ is the dot-product operator; $\|\mathbf{z}_q\|^2$ is the squared Euclidean norm of the position of $\mathbf{z}_q$; VN-In($\cdot$) is an invariant VN network⁴⁴ that converts the equivariant shape embedding $\mathbf{H}^s \in \mathbb{R}^{d_p \times 3}$ into an invariant shape embedding $\text{VN-In}(\mathbf{H}^s) \in \mathbb{R}^{d_p}$. Intuitively, SE-dec predicts the signed distance between the query point and 3D surface by jointly considering the interaction between the point and surface ($\langle \mathbf{z}_q, \mathbf{H}^s \rangle$), the distance of the query point ($\|\mathbf{z}_q\|^2 = \langle \mathbf{z}_q, \mathbf{z}_q \rangle$) to the origin, and the molecule surface shape (VN-In($\cdot$)). All these three terms are invariant to any rotation transformations, as they are calculated from the dot-product operation $\langle \cdot, \cdot \rangle$. This operation is invariant to any rotations as $\langle \mathbf{R}\mathbf{z}, \mathbf{R}\mathbf{z} \rangle = \mathbf{z}^T \mathbf{R}^T \mathbf{R} \mathbf{z} = \mathbf{z}^T \mathbf{z} = \langle \mathbf{z}, \mathbf{z} \rangle$. Note that VN-In($\cdot$) comprises invariant dot-product operations and specifically designed invariant activations to learn invariant embeddings, as detailed by Deng et al.⁴⁴. The predicted signed distance $\hat{o}_q$ is used to calculate the loss for the optimization of $\mathbf{H}^s$ (shown in equation (3)). We present the sampling process of $\mathbf{z}_q$ in Supplementary Section 15.

SE pretraining. DiffSMol pretrains SE by minimizing the squared-error loss between the predicted and the ground-truth signed distances of query points to the surface shape $\mathcal{S}$ as follows:

$$\mathcal{L}^s = \sum_{\mathbf{z}_q \in \mathcal{Z}} \|\hat{o}_q - o_q\|^2, \quad (3)$$

where $\mathcal{Z}$ is the set of sampled query points and o_q is the ground-truth signed distance of query point $\mathbf{z}_q$. By pretraining SE, DiffSMol learns $\mathbf{H}^s$ that will be used as the condition in the following 3D molecule generation.

Diffusion-based molecule generation

This section is largely based on our preliminary work²⁰. In DiffSMol, a shape-conditioned molecule diffusion model, referred to as DIFF, is used to generate a 3D molecule structure (that is, atom coordinates and features, and bonds) conditioned on a given 3D surface shape that is represented by the shape latent embedding $\mathbf{H}^s$ (equation (1)). Figure 1b shows the architecture of DIFF. Following the denoising diffusion probabilistic models⁴⁶, DIFF includes a forward diffusion process based on a Markov chain, denoted as DIFF-forward, which gradually adds noises step by step to the atom positions and features $\{\mathbf{x}_t, \mathbf{v}_t\}$ in the training molecules with i indexing the i th atom. The noisy atom positions and features at step t are represented as $\{\mathbf{x}_{i,t}, \mathbf{v}_{i,t}\}$ ($t = 1, \dots, T$), and the molecules without any noise are represented as $\{\mathbf{x}_{i,0}, \mathbf{v}_{i,0}\}$. At the final step T , $\{\mathbf{x}_{i,T}, \mathbf{v}_{i,T}\}$ are completely unstructured and resemble a simple distribution like a normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$ or a uniform categorical distribution $\mathcal{C}(\mathbf{I}/K)$, in which $\mathbf{I}$ and $\mathbf{I}$ denotes the identity matrix and identity vector, respectively. When no ambiguity arises, we will eliminate subscript i in the notations and use $(\mathbf{x}_t, \mathbf{v}_t)$ for brevity.

During training, DIFF is learned to reverse the forward diffusion process via another Markov chain, referred to as the backward generative process and denoted as DIFF-backward, to remove the noises in the noisy molecules. During inference, DIFF first samples noisy atom positions and features at step T from simple distributions and then generates a 3D molecule structure by removing the noises in the noisy molecules step by step until t reaches 1.

Forward diffusion process. Following the previous work²³, at step $t \in [1, T]$, a small Gaussian noise and a small categorical noise are added to the continuous atom positions and discrete atom features $\{\mathbf{x}_{t-1}, \mathbf{v}_{t-1}\}$, respectively. The noise levels of the Gaussian and categorical noises are determined by two predefined variance schedules $(\beta_t^x, \beta_t^v) \in (0, 1)$, where β_t^x and β_t^v are selected to be sufficiently small to ensure the smoothness of DIFF-forward. The details about the entire forward process are presented in Supplementary Section 16.

Backward generative process. DIFF learns to reverse DIFF-forward by denoising from $(\mathbf{x}_t, \mathbf{v}_t)$ to $(\mathbf{x}_{t-1}, \mathbf{v}_{t-1})$ at $t \in [1, T]$, conditioned on the shape latent embedding $\mathbf{H}^s$. Specifically, the probabilities of $(\mathbf{x}_{t-1}, \mathbf{v}_{t-1})$ denoised from $(\mathbf{x}_t, \mathbf{v}_t)$ are estimated by the approximates of the ground-truth posteriors $p(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ (supplementary equation (8) in Supplementary Section 16.1) and $p(\mathbf{v}_{t-1}|\mathbf{v}_t, \mathbf{v}_0)$ (supplementary equation (10) in Supplementary Section 16.1). Given that $(\mathbf{x}_0, \mathbf{v}_0)$ is unknown in the generative process, a prediction module SMP, which is a GNN with multiple layers (see ‘Shape-conditioned molecule prediction’ section) is used to predict the atom position and feature $(\mathbf{x}_0, \mathbf{v}_0)$ at time step t as

$$(\hat{\mathbf{x}}_{0,t}, \hat{\mathbf{v}}_{0,t}) = \text{SMP}(\mathbf{x}_t, \mathbf{v}_t, \mathbf{H}^s), \quad (4)$$

where $\hat{\mathbf{x}}_{0,t}$ and $\hat{\mathbf{v}}_{0,t}$ are the predictions of $\mathbf{x}_0$ and $\mathbf{v}_0$ based on the information at t (that is, $\mathbf{x}_t, \mathbf{v}_t$ and $\mathbf{H}^s$). More details are available in Supplementary Section 17.

Model training. DiffSMol optimizes DIFF by minimizing the following three losses.

Atom position loss. DiffSMol measures the squared errors between the predicted positions ($\mathbf{x}_{0,t}$) from the prediction module SMP (equation (4)) and the ground-truth positions ($\mathbf{x}_0$) of atoms in molecules. Given a particular step t , the loss is calculated as

$$\mathcal{L}_t^x(\mathcal{M}) = w_t^x \sum_{\forall a \in \mathcal{M}} \|\mathbf{x}_{0,t} - \mathbf{x}_0\|^2, \quad (5)$$

where $w_t^x = \min(\lambda_t, \delta)$, $\lambda_t = \bar{\alpha}_t^x / (1 - \bar{\alpha}_t^x)$,

where w_t^x is a weight at step t and is calculated by clipping the signal-to-noise ratio $\lambda_t > 0$ with a threshold $\delta > 0$. Note that, because $\bar{\alpha}_t^x$ decreases monotonically as t increases from 1 to T (supplementary equation (7) in Supplementary Section 16.1), w_t^x decreases monotonically over t as well until it is clipped. Thus, w_t^x imposes lower weights on the loss when the noise level in $\mathbf{x}_t$ is higher (that is, t close to T). This encourages the model training to focus more on accurately recovering molecule structures when there are sufficient signals in the data, rather than being potentially confused by major noises in the data.

Atom feature loss. DiffSMol also minimizes the Kullback–Leibler divergence (KL) divergence⁴⁷ between the ground-truth posterior $p(\mathbf{v}_{t-1}|\mathbf{v}_t, \mathbf{v}_0)$ (supplementary equation (10) in Supplementary Section 16.1) and its approximate $p_\theta(\mathbf{v}_{t-1}|\mathbf{v}_t, \bar{\mathbf{v}}_{0,t})$ (supplementary equation (22) in Supplementary Section 17) for discrete atom features to optimize DIFF, following the literature⁴⁸. In particular, the KL divergence at t for a given molecule $\mathcal{M}$ is calculated as

$$\begin{aligned} \mathcal{L}_t^v(\mathcal{M}) &= \sum_{\forall a \in \mathcal{M}} \text{KL}(p(\mathbf{v}_{t-1}|\mathbf{v}_t, \mathbf{v}_0) \| p_\theta(\mathbf{v}_{t-1}|\mathbf{v}_t, \bar{\mathbf{v}}_{0,t})), \\ &= \sum_{\forall a \in \mathcal{M}} \text{KL}(\mathbf{c}(\mathbf{v}_t, \mathbf{v}_0) \| \mathbf{c}_\theta(\mathbf{v}_t, \bar{\mathbf{v}}_{0,t})), \end{aligned} \quad (6)$$

where $\mathbf{c}(\mathbf{v}_t, \mathbf{v}_0)$ is a categorical distribution of $\mathbf{v}_{t-1}$ (supplementary equation (22)); $\mathbf{c}_\theta(\mathbf{v}_t, \bar{\mathbf{v}}_{0,t})$ is an estimate of $\mathbf{c}(\mathbf{v}_t, \mathbf{v}_0)$ (supplementary equation (22)).

Bond type loss. DiffSMol also minimizes the classification errors between the predicted bond types ($\mathbf{e}_{ij}$) and the ground-truth types $\mathbf{b}_{ij}$ of bonds in molecules. At step t , for each l th layer of SMP, DiffSMol predicts the bond types ($\mathbf{e}_{ij,t,l}$) to understand the relations among atoms. Details about the calculation of $\mathbf{e}_{ij,t,l}$ will be discussed later in equation (20). Given a particular step t , the error on bond type prediction at the l th layer is calculated as

$$\mathcal{L}_{t,l}^b(\mathcal{M}) = \sum_{\forall a_i \in \mathcal{M}} \sum_{\forall a_j \in N(\mathbf{x}_{i,t})} H(\mathbf{e}_{ij,t,l}, \mathbf{b}_{ij}), \quad (7)$$

where $N(\mathbf{x}_{i,t})$ denotes the k -nearest neighbours of atom a_i in position $\mathbf{x}_{i,t}$; and $H(\cdot)$ denotes the cross-entropy loss. The bond type prediction loss across different layers is then aggregated as

$$\mathcal{L}_t^b(\mathcal{M}) = \frac{w_t^x}{L-1} \sum_{l=1}^{L-1} \mathcal{L}_{t,l}^b(\mathcal{M}) + w_t^x \mathcal{L}_{t,L}^b(\mathcal{M}), \quad (8)$$

where w_t^x is the weight at step t used in equation (5); L is the number of layers in SMP. As in equation (5), the w_t^x in equation (8) is used to encourage the model training to focus more on accurately predicting bond types when the data provide sufficient signals, rather than being confused by major noises in the data. Note that, similar to Jumper et al.⁴⁹, in equation (8), DiffSMol uses different weights on the last layer (that is, $l = L$) and all the other layers, as we empirically find that this design benefits the generation performance.

Overall DiffSMol loss. The overall DiffSMol loss function is defined as

$$\mathcal{L} = \sum_{\forall \mathcal{M} \in \mathcal{M}} \sum_{\forall t \in \mathcal{T}} (\mathcal{L}_t^x(\mathcal{M}) + \xi \mathcal{L}_t^v(\mathcal{M}) + \zeta \mathcal{L}_t^b(\mathcal{M})), \quad (9)$$

where $\mathcal{M}$ is the set of all the molecules in training; $\mathcal{T}$ is the set of timesteps; $\xi > 0$ and $\zeta > 0$ are two hyperparameters to balance $\mathcal{L}_t^x(\mathcal{M})$, $\mathcal{L}_t^v(\mathcal{M})$ and $\mathcal{L}_t^b(\mathcal{M})$. During training, step t is uniformly sampled from $\mathcal{T} = \{1, 2, \dots, 1000\}$. The derivation of the loss functions is available in Supplementary Section 18.

Shape-conditioned molecule prediction

In DIFF-backward, the prediction module SMP (equation (4)) predicts the atom positions and features ($\mathbf{x}_{0,t}, \bar{\mathbf{v}}_{0,t}$) given the noisy data ($\mathbf{x}_t, \mathbf{v}_t$) conditioned on $\mathbf{H}^s$ (ref. 20). For brevity, in this section, we eliminate the subscript t in the notations when no ambiguity arises. In particular, as shown in Fig. 1d, SMP is a multilayer GNN comprising L layers. In the l th layer, SMP uses the geometric vector perceptron (GVP) to learn a scalar embedding $\mathbf{a}_{i,l} \in \mathbb{R}^{d_a}$ and a vector embedding $\mathbf{r}_i \in \mathbb{R}^{3 \times d_r}$ for atom a_i in an alternative manner that guarantees the invariance of $\mathbf{a}_{i,l}$ and the equivariance of $\mathbf{r}_{i,l}$ (ref. 50). Intuitively, $\mathbf{a}_{i,l}$ captures inherent properties (for example, atom types) of atom a_i , which are invariant of the molecule's orientation or position in 3D space. Different from $\mathbf{a}_{i,l}$, $\mathbf{r}_{i,l}$ captures geometric information (for example, atom positions) of atom a_i , which will change under different transformations. We note that existing work primarily uses equivariant GNNs (EGNNs)⁵¹ for the prediction. However, EGNNs could suffer from limited capacity in capturing rich geometric information within molecules as they can only represent geometric features in a 3D latent space. By contrast, GVP exhibits stronger expressiveness, capable of learning latent embeddings in spaces of any dimensions⁵². Equipped with GVP, SMP enables the learning of effective representations for geometric information. Ablation studies comparing EGNN and GVP for both SMG and PMG in Supplementary Section 19 also demonstrate the advantages of using GVP in SMP. Note that, to ensure translation equivariance, SMP shifts a fixed point (that is, the centre of shape condition $\mathcal{P}$) to zero to eliminate all translations. Therefore, only rotation equivariance needs to be considered.

SMP also leverages shape-aware scalar embeddings $\hat{\mathbf{a}}_{i,l}$ and vector embeddings $\hat{\mathbf{r}}_{i,l}$ to generate molecules tailored to the shape condition. SMP learns $\hat{\mathbf{a}}_{i,l}$ from $\mathbf{a}_{i,l}$ using the shape representation $\mathbf{H}^s$ in an invariant way (equation (17)). Similarly, SMP learns $\hat{\mathbf{r}}_{i,l}$ from $\mathbf{r}_{i,l}$ and $\mathbf{H}^s$ in an equivariant manner (equation (19)). In addition, SMP utilizes the bond type embeddings to enhance the understanding of relations among atoms for better prediction (equation (20)).

In particular, SMP estimates the type and position of the i th atom a_i as

$$\bar{\mathbf{x}}_{0,i} = \mathbf{x}_i + \mathbf{r}_{i,L}, \quad \bar{\mathbf{v}}_{0,i} = \text{softmax}(\text{MLP}(\mathbf{a}_{i,L})), \quad (10)$$

where $\bar{\mathbf{v}}_{0,i}$ (equation (4)) is the predicted probability distribution across all the types of atom features; $\bar{\mathbf{x}}_{0,i}$ (equation (4)) is the predicted position of a_i ; $\mathbf{x}_i$ is the noisy position of a_i ; and $\mathbf{a}_{i,L}$ and $\mathbf{r}_{i,L}$ are the invariant scalar embedding and equivariant vector embedding for atom a_i , respectively. In each l th layer, $\mathbf{a}_{i,l}$ and $\mathbf{r}_{i,l}$ of atom a_i are updated by propagating its neighbourhood's inherent features and geometric features as

$$\mathbf{a}_{i,l}, \mathbf{r}_{i,l} = \text{GVP}(\mathbf{h}_{i,l}, \mathbf{y}_{i,l}), \quad (11)$$

$$\mathbf{h}_{i,l} = \left[\mathbf{v}_i, \hat{\mathbf{a}}_{i,l-1}, \sum_{j \in N(i)} e_{ji,l} \mathbf{m}_{ji,l}, t \right], \quad \mathbf{y}_{i,l} = \left[\mathbf{x}_i, \hat{\mathbf{r}}_{i,l-1}, \sum_{j \in N(i)} e_{ji,l} \mathbf{n}_{ji,l} \right], \quad (12)$$

$$\hat{\mathbf{a}}_{i,l-1}, \hat{\mathbf{r}}_{i,l-1} = \text{SARL}(\mathbf{a}_{i,l-1}, \mathbf{r}_{i,l-1}, \mathbf{H}^s), \quad (13)$$

where $\text{GVP}(\cdot)$ is a function that learns $\mathbf{a}_{i,l}$ and $\mathbf{r}_{i,l}$ jointly from $\mathbf{h}_{i,l} \in \mathbb{R}^{d_h}$ and $\mathbf{y}_{i,l} \in \mathbb{R}^{3 \times d_y}$; $[\cdot, \cdot]$ is the concatenation operation; $N(i)$ denotes the k -nearest neighbour atoms of atom a_i over the 3D space; t denotes the time step; $\mathbf{v}_i$ is the noisy feature vector of a_i ; $\mathbf{m}_{ji,l} \in \mathbb{R}^{d_m}$ and $\mathbf{n}_{ji,l} \in \mathbb{R}^{3 \times d_n}$

are messages to propagate information from a_j to a_i as will be described in equation (16); $e_{ji,l}$ is the attention weight used to aggregate information from neighbouring atoms; shape-aware atom representation learning (SARL) is a module to learn shape-aware atom embeddings as will be introduced later; $\hat{\mathbf{a}}_{i,l-1}$ and $\hat{\mathbf{r}}_{i,l-1}$ are the shape-aware atom scalar and vector embedding, respectively (detailed in equations (17) and (19)). The weight $e_{ji,l}$ is calculated to estimate how much the neighbouring atom a_j should contribute to the learning of $\mathbf{h}_{i,l}$ and $\mathbf{y}_{i,l}$ as follows:

$$e_{ji,l} = \frac{\exp(Q_{i,l}K_{ji,l})}{\sum_{k \in \mathcal{N}(i)} \exp(Q_{i,l}K_{ki,l})},$$

$$\text{where } Q_{i,l} = \text{MLP}([\hat{\mathbf{a}}_{i,l-1}, \|\hat{\mathbf{r}}_{i,l-1}\|^2]),$$

$$K_{ji,l} = \text{MLP}([\mathbf{m}_{ji,l}, \|\mathbf{n}_{ji,l}\|^2]).$$
(14)

In both equations (12) and (14), the messages $\mathbf{m}_{ji,l}$ and $\mathbf{n}_{ji,l}$ are calculated from the scalar embeddings (for example, $\hat{\mathbf{a}}_{i,l}$) and vector embeddings (for example, $\hat{\mathbf{r}}_{i,l}$) of atoms as

$$\mathbf{m}_{ji,l}, \mathbf{n}_{ji,l} = \text{GVP}(\hat{\mathbf{m}}_{ji,l}, \hat{\mathbf{n}}_{ji,l}),$$
(15)

$$\hat{\mathbf{m}}_{ji,l} = [\hat{\mathbf{a}}_{j,l-1}, d_{ji}, \mathbf{e}_{ji,l-1}], \quad \hat{\mathbf{n}}_{ji,l} = [\hat{\mathbf{r}}_{j,l-1}, \mathbf{x}_j - \mathbf{x}_i],$$
(16)

where $\text{GVP}(\cdot)$ is a function that learns $\mathbf{m}_{ji,l}$ and $\mathbf{n}_{ji,l}$ jointly from $\hat{\mathbf{m}}_{ji,l} \in \mathbb{R}^{d_m}$ and $\hat{\mathbf{n}}_{ji,l} \in \mathbb{R}^{3 \times d_v}$; $[\cdot, \cdot]$ is the concatenation operation; $\mathbf{e}_{ji,l-1}$ is the embedding of the bond type between a_i and a_j (detailed in equation (20)); and d_{ji} is the distance between $\mathbf{x}_i$ and $\mathbf{x}_j$.

Shape-aware atom representation learning. To generate molecules that tailored to the shape condition represented by $\mathbf{H}^s$, SMP adapts the scalar embedding $\mathbf{a}_{i,l}$ and the vector embedding $\mathbf{r}_{i,l}$ of each atom a_i into the shape-aware scalar embedding $\hat{\mathbf{a}}_{i,l}$ and the shape-aware vector embedding $\hat{\mathbf{r}}_{i,l}$ by incorporating $\mathbf{H}^s$ at each layer. In particular, SMP learns $\hat{\mathbf{a}}_{i,l}$ for each atom a_i using $\mathbf{H}^s$ as follows:

$$\hat{\mathbf{a}}_{i,l} = \text{MLP}([\mathbf{a}_{i,l}, \mathbf{o}_{i,l}]),$$
(17)

where $[\cdot, \cdot]$ is the concatenation operation; $\mathbf{a}_{i,l}$ is the scalar embedding of atom a_i at the l th layer; and $\mathbf{o}_{i,l} \in \mathbb{R}^{d_o}$ represents the inherent relations between a_i and the molecular surface shape, such as the signed distance from a_i to the shape. SMP learns $\mathbf{o}_{i,l}$ in a similar way to equation (2) as follows:

$$\mathbf{o}_{i,l} = \text{MLP}([\mathbf{a}_{i,l}, \langle \mathbf{r}_{i,l}, \mathbf{H}^s \rangle, \|\mathbf{r}_{i,l}\|, \text{VN-In}(\mathbf{H}^s)]),$$
(18)

where $\langle \mathbf{r}_{i,l}, \mathbf{H}^s \rangle$ is the dot product between $\mathbf{r}_{i,l}$ and $\mathbf{H}^s$; $\|\mathbf{r}_{i,l}\|^2$ is the column-wise Euclidean norm of the vector feature $\mathbf{r}_{i,l}$; and $\text{VN-In}(\mathbf{H}^s)$ encodes the inherent geometry of shape condition and thus is shared across all the layers. Apart from scalar embeddings, SMP also incorporates shape information into the vector embeddings as follows:

$$\hat{\mathbf{r}}_{i,l} = \text{VN-MLP}([\mathbf{r}_{i,l}, \mathbf{H}^s]),$$
(19)

where $\mathbf{r}_{i,l}$ is the vector embedding of atom a_i at the l th layer; $\text{VN-MLP}(\cdot)$ is an equivariant VN network⁴⁴ that learns nonlinear interactions $\hat{\mathbf{r}}_{i,l} \in \mathbb{R}^{3 \times d_r}$ between $\mathbf{r}_{i,l}$ and $\mathbf{H}^s$ in an equivariant way. Ablation studies on SARL for both SMG and PMG in Supplementary Section 19 demonstrate that DiffSMol equipped with SARL is more effective at generating molecules with highly similar shapes to condition molecules.

Bond-type representation learning. As shown in equation (16), SMP leverages the types of bonds within M to facilitate its understanding of relations among atoms. In particular, for the bond between a_j and a_i , SMP generates the bond type embedding as follows:

$$\mathbf{e}_{ji,l} = \begin{cases} \text{MLP}([\mathbf{a}_{i,l} + \mathbf{a}_{j,l}, \text{abs}(\mathbf{a}_{i,l} - \mathbf{a}_{j,l}), d_{ji}]), & \text{if } l = 0, \\ \text{MLP}([\mathbf{a}_{i,l} + \mathbf{a}_{j,l}, \text{abs}(\mathbf{a}_{i,l} - \mathbf{a}_{j,l}), \|\mathbf{r}_i\|^2 \\ + \|\mathbf{r}_j\|^2, \text{abs}(\|\mathbf{r}_i\|^2 - \|\mathbf{r}_j\|^2)]), & \text{if } l > 0, \end{cases} \quad (20)$$

where $\mathbf{a}_{i,l}$ and $\mathbf{r}_{i,l}$ is the scalar embedding and vector embedding of a_i (equation (12)), respectively; $\text{abs}(\cdot)$ represents the absolute difference; and d_{ji} is the distance between the positions $\mathbf{x}_j$ and $\mathbf{x}_i$. SMP guarantees that the predictions $\mathbf{e}_{ij,l}$ and $\mathbf{e}_{ji,l}$ are invariant to the permutation of atom a_i and a_j . This is achieved by using two invariant operations: the sum and the absolute difference operation. To learn effective bond-type embeddings, we also use the sum and the absolute difference of column-wise Euclidean norm of $\mathbf{r}_i^l$ and $\mathbf{r}_j^l$ to implicitly estimate the distance between a_i and a_j . When $l = 0$, we directly use the distance d_{ji} to calculate $\mathbf{e}_{ji,l}$. Ablation studies on bond-type representation learning (BTRL) for both SMG and PMG in Supplementary Section 19 suggest that incorporating BTRL substantially improves the capability of DiffSMol in generating stable and high-quality molecules.

Guidance-induced inference

During inference, DiffSMol generates molecules by gradually denoising $(\mathbf{x}_T, \mathbf{v}_T)$ to $(\mathbf{x}_0, \mathbf{v}_0)$ using the prediction module SMP. Specifically, DiffSMol samples $\mathbf{x}_T$ and $\mathbf{v}_T$ from $\mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\mathcal{C}(\mathbf{1}/K)$, respectively. After that, DiffSMol samples $\mathbf{x}_{t-1}$ from $\mathbf{x}_t$ using $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0, \mathbf{v}_t)$ (supplementary equation (21)). Similarly, DiffSMol samples $\mathbf{v}_{t-1}$ from $\mathbf{v}_t$ using $p_{\theta}(\mathbf{v}_{t-1}|\mathbf{v}_t, \mathbf{v}_0, \mathbf{x}_t)$ (supplementary equation (22)) until t reaches 1. DiffSMol uses postprocessing to determine the bond type between atoms based on atomic distances following the previous work^{22,23}. Although the learned bond type embeddings in DiffSMol could provide valuable topology information for molecule prediction (SMP), we observe that directly using predicted bond types in generated molecules could lead to suboptimal performance.

DiffSMol with shape guidance. This section is largely based on our preliminary work²⁰. During molecule generation, as shown in Fig. 1c, DiffSMol can also utilize additional shape guidance by pushing the predicted atoms to the shape of the condition molecule M_x . This approach is motivated by previous work⁵³, which demonstrates that incorporating additional guidance into conditional diffusion models can further ensure the generated objects closely following the given condition. Note that, different from the shape for conditions, when used as guidance, we define molecule shapes as a set of points $\mathcal{Q}$ sampled according to atom positions in the condition molecule M_x following Adams and Coley et al.¹⁹. We empirically find that this design leads to improved generation performance. In particular, for each atom a_i in M_x , 20 points are randomly sampled into $\mathcal{Q}$ from a Gaussian distribution centred at $\mathbf{x}_i$. Given the predicted atom position $\mathbf{x}_{0,t}$ at step t , DiffSMol applies the shape guidance by adjusting the predicted positions to $\mathcal{Q}$ as follows:

$$\mathbf{x}_{0,t}^* = (1 - \sigma)\mathbf{x}_{0,t} + \sigma \sum_{\mathbf{z} \in \mathcal{N}(\mathbf{x}_{0,t}; \mathcal{Q})} \mathbf{z}/k, \text{ when } \sum_{\mathbf{z} \in \mathcal{N}(\mathbf{x}_{0,t}; \mathcal{Q})} d(\mathbf{x}_{0,t}, \mathbf{z})/k > \gamma, \quad (21)$$

where $\sigma > 0$ is the weight used to balance the prediction $\mathbf{x}_{0,t}$ and the adjustment; $d(\mathbf{x}_{0,t}, \mathbf{z})$ is the Euclidean distance between $\mathbf{x}_{0,t}$ and $\mathbf{z}$; $\mathcal{N}(\mathbf{x}_{0,t}; \mathcal{Q})$ is the set of k -nearest neighbours of $\mathbf{x}_{0,t}$ in $\mathcal{Q}$ based on $d(\cdot)$; and $\gamma > 0$ is a distance threshold. By doing the above adjustment, the predicted atom positions will be pushed to those of M_x if they are sufficiently far away. Note that the shape guidance is applied exclusively for steps

$$t = T, T-1, \dots, S, \text{ where } S > 1, \quad (22)$$

not for all the steps, and thus it adjusts predicted atom positions only when there is a lot of noise and the prediction needs more guidance. DiffSMol with the shape guidance is referred to as DiffSMol+s.

DiffSMol with protein pocket guidance. When applying DiffSMol to PMG (that is, protein pocket of the condition molecule is available), we observe that atoms in the generated molecules could be too close to the protein pocket atoms $\mathcal{K}$, thereby leading to steric clashes and, thus, undesirable binding affinities. To address this issue, as shown in Fig. 1e, DiffSMol utilizes pocket guidance to further adjust atom positions and maintain sufficient distances between molecule atoms and protein atoms. In particular, DiffSMol refines the atom positions in the generated molecules based on $\mathcal{K}$ as follows:

$$\mathbf{x}_t^* = \mathbf{x}_t + \frac{\mathbf{x}_t - \mathbf{z}}{d(\mathbf{x}_t, \mathbf{z})} \times (\rho - d(\mathbf{x}_t, \mathbf{z}) + \epsilon) \quad \text{if } \exists \mathbf{z} \in N(\mathbf{x}_t; \mathcal{K}), d(\mathbf{x}_t, \mathbf{z}) < \rho, \quad (23)$$

where $\mathbf{x}_t$ is the sampled atom positions at the step t ; $N(\mathbf{x}_t; \mathcal{K})$ is the set of k -nearest neighbours of $\mathbf{x}_t$ within the protein atoms $\mathcal{K}$; $d(\mathbf{x}_t, \mathbf{z})$ is the distance between $\mathbf{x}_t$ and $\mathbf{z}$; and $\frac{\mathbf{x}_t - \mathbf{z}}{d(\mathbf{x}_t, \mathbf{z})}$ calculates the unit vector in the direction that moves $\mathbf{x}_t$ far from $\mathbf{z}$. DiffSMol introduces a threshold ρ to assess if protein atoms and molecule atoms are too close. DiffSMol identifies this threshold from known protein–ligand complexes in the training dataset. DiffSMol also introduces a hyperparameter ϵ to control the margin. Note that, different from the shape guidance that is applied on $\mathbf{x}_{0:t}$, the pocket guidance is applied on $\mathbf{x}_t$. We empirically find that this design benefits the generated molecules in their binding affinities to protein pockets. DiffSMol with the pocket guidance is referred to as DiffSMol+p.

Data availability

The MOSES dataset is available via GitHub at <https://github.com/molecularsets/moses>, and the CrossDocked2020 dataset is available via GitHub at <https://github.com/gnina/models/tree/master/data/CrossDocked2020>. Additional data, including our generated molecules and trained models, are publicly available via GitHub at <https://github.com/ninglab/DiffSMol>.

Code availability

The code for DiffSMol is publicly available via GitHub at <https://github.com/ninglab/DiffSMol>.

References

- Sun, D., Gao, W., Hu, H. & Zhou, S. Why 90% of clinical drug development fails and how to improve it? *Acta Pharm. Sin. B* **12**, 3049–3062 (2022).
- Wouters, O. J., McKee, M. & Luyten, J. Estimated research and development investment needed to bring a new medicine to market, 2009–2018. *JAMA* **323**, 844–853 (2020).
- Yu, W. & Mackerell, A. D. *Computer-Aided Drug Design Methods* 85–106 (Springer, 2016).
- Acharya, C., Coop, A., Polli, J. E. & Mackerell, A. D. Recent advances in ligand-based drug design: relevance and utility of the conformationally sampled pharmacophore approach. *Curr. Comput. Aided Drug Des.* **7**, 10–22 (2011).
- Anderson, A. C. The process of structure-based drug design. *Chem. Biol.* **10**, 787–797 (2003).
- Gimeno, A. et al. The light and dark sides of virtual screening: what is there to know? *Int. J. Mol. Sci.* **20**, 1375 (2019).
- Kingma, D. P. & Welling, M. Auto-encoding variational Bayes. In *Proc. 2nd International Conference on Learning Representations* (eds Bengio, Y. & LeCun, Y.) (OpenReview.net, 2014).
- Song, J., Meng, C. & Ermon, S. Denoising diffusion implicit models. In *Proc. 9th International Conference on Learning Representations* (eds Oh, A. et al.) (OpenReview.net, 2021).
- OpenAI et al. GPT-4 technical report. Preprint at <https://arxiv.org/abs/2303.08774> (2023).
- Yu, B., Baker, F. N., Chen, Z., Ning, X. & Sun, H. LLaSMol: advancing large language models for chemistry with a large-scale, comprehensive, high-quality instruction tuning dataset. In *Proc. First Conference on Language Modeling* (OpenReview.net, 2024). <https://openreview.net/forum?id=IY6XTF9tPv>
- Jin, W., Barzilay, R. & Jaakkola, T. Junction tree variational autoencoder for molecular graph generation. In *Proc. 35th International Conference on Machine Learning* Vol. 80 (eds Dy, J. & Krause, A.) 2323–2332 (PMLR, 2018).
- Schneuing, A. et al. Structure-based drug design with equivariant diffusion models. *Nat. Comput. Sci.* <https://doi.org/10.1038/s43588-024-00737-x> (2024).
- Liu, S. et al. Conversational drug editing using retrieval and domain feedback. In *Proc. 12th International Conference on Learning Representations* (eds Chaudhuri, S. et al.) (OpenReview.net, 2024).
- Boström, J., Hogner, A. & Schmitt, S. Do structurally similar ligands bind in a similar fashion? *J. Med. Chem.* **49**, 6716–6725 (2006).
- Gómez-Bombarelli, R. et al. Automatic chemical design using a data-driven continuous representation of molecules. *ACS Cent. Sci.* **4**, 268–276 (2018).
- Chen, Z., Min, M. R., Parthasarathy, S. & Ning, X. A deep generative model for molecule optimization via one fragment modification. *Nat. Mach. Intell.* **3**, 1040–1049 (2021).
- Hoogeboom, E., Satorras, V. G., Vignac, C. & Welling, M. Equivariant diffusion for molecule generation in 3D. In *Proc. 39th International Conference on Machine Learning* Vol. 162 (eds Chaudhuri, K. et al.) 8867–8887 (PMLR, 2022).
- Long, S., Zhou, Y., Dai, X. & Zhou, H. Zero-shot 3d drug design by sketching and generating. In *Proc. 36th International Conference on Neural Information Processing Systems* (eds Oh, A. H. et al.) 23894–23907 (Curran Associates, 2022).
- Adams, K. & Coley, C. W. Equivariant shape-conditioned generation of 3D molecules for ligand-based drug design. In *Proc. 11th International Conference on Learning Representations* (eds Nickel, M. et al.) (OpenReview.net, 2023).
- Chen, Z., Peng, B., Parthasarathy, S. & Ning, X. Shape-conditioned 3D molecule generation via equivariant diffusion models. In *Proc. NeurIPS 2023 Generative AI and Biology (GenBio) Workshop* (2023). <https://openreview.net/forum?id=JWfvMT43pZ>
- Luo, S., Guan, J., Ma, J. & Peng, J. A. 3D generative model for structure-based drug design. In *Proc. 35th International Conference on Neural Information Processing Systems* (eds Beygelzimer, A. et al.) 6229–6239 (Curran Associates, 2021).
- Peng, X. et al. Pocket2Mol: efficient molecular sampling based on 3D protein pockets. In *Proc. 39th International Conference on Machine Learning* Vol. 162 (eds Chaudhuri, K. et al.) 17644–17655 (PMLR, 2022).
- Guan, J. et al. 3D equivariant diffusion for target-aware molecule generation and affinity prediction. In *Proc. 11th International Conference on Learning Representations* (eds Nickel, M. et al.) (OpenReview.net, 2023).
- Tingle, B. I. et al. Zinc-22—a free multi-billion-scale database of tangible compounds for ligand discovery. *J. Chem. Inf. Model.* **63**, 1166–1176 (2023).
- Polykovskiy, D. et al. Molecular Sets (MOSES): a benchmarking platform for molecular generation models. *Front. Pharmacol.* **11**, 565644 (2020).
- Guan, J. et al. DecompDiff: diffusion models with decomposed priors for structure-based drug design. In *Proc. 40th International Conference on Machine Learning* Vol. 202 (eds Krause, A. et al.) 11827–11846 (PMLR, 2023).
- Ferreira, L., dos Santos, R., Oliva, G. & Andricopulo, A. Molecular docking and structure-based drug design strategies. *Molecules* **20**, 13384–13421 (2015).

28. Tadesse, S., Yu, M., Kumarasiri, M., Le, B. T. & Wang, S. Targeting CDK6 in cancer: state of the art and new insights. *Cell Cycle* **14**, 3220–3230 (2015).
29. El-Amouri, S. S. et al. Neprilysin: an enzyme candidate to slow the progression of Alzheimer's disease. *Am. J. Pathol.* **172**, 1342–1354 (2008).
30. Burley, S. K. et al. RCSB protein data bank (rcsb.org): delivery of experimentally-determined PDB structures alongside one million computed structure models of proteins from artificial intelligence/machine learning. *Nucleic Acids Res.* **51**, D488–D508 (2022).
31. Bickerton, G. R., Paolini, G. V., Besnard, J., Muresan, S. & Hopkins, A. L. Quantifying the chemical beauty of drugs. *Nat. Chem.* **4**, 90–98 (2012).
32. Ertl, P. & Schuffenhauer, A. Estimation of synthetic accessibility score of drug-like molecules based on molecular complexity and fragment contributions. *J. Cheminform.* **1**, 8 (2009).
33. Neves, M. A. C., Totrov, M. & Abagyan, R. Docking and scoring with icm: the benchmarking results and strategies for improvement. *J. Comput. Aided Mol. Des.* **26**, 675–686 (2012).
34. Yang, H. et al. admetSAR 2.0: web-service for prediction and optimization of chemical admet properties. *Bioinformatics* **35**, 1067–1069 (2018).
35. Halgren, T. A. Merck molecular force field. I. Basis, form, scope, parameterization, and performance of mmff94. *J. Comput. Chem.* **17**, 490–519 (1996).
36. Lipinski, C. A., Lombardo, F., Dominy, B. W. & Feeney, P. J. Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. *Adv. Drug Deliv. Rev.* **23**, 3–25 (1997).
37. Patnaik, A. et al. Efficacy and safety of abemaciclib, an inhibitor of CDK4 and CDK6, for patients with breast cancer, non-small cell lung cancer, and other solid tumors. *Cancer Discov.* **6**, 740–753 (2016).
38. Lu, J. Palbociclib: a first-in-class CDK4/CDK6 inhibitor for the treatment of hormone-receptor positive advanced breast cancer. *J. Hematol. Oncol.* **8**, 98 (2015).
39. Tripathy, D., Bardia, A. & Sellers, W. R. Ribociclib (lee011): mechanism of action and clinical impact of this selective cyclin-dependent kinase 4/6 inhibitor in various solid tumors. *Clin. Cancer Res.* **23**, 3251–3262 (2017).
40. Benigni, R., Bossa, C., Tcheremenskaia, O. & Giuliani, A. Alternatives to the carcinogenicity bioassay: in silico methods, and their in vitro and in vivo mutagenicity assays. *Expert Opin. Drug Metab. Toxicol.* **6**, 809–819 (2010).
41. Soo, J. Y.-C., Jansen, J., Masereeuw, R. & Little, M. H. Advances in predictive in vitro models of drug-induced nephrotoxicity. *Nat. Rev. Nephrol.* **14**, 378–393 (2018).
42. Du, X. et al. Insights into protein–ligand interactions: mechanisms, models, and methods. *Int. J. Mol. Sci.* **17**, 144 (2016).
43. Park, J. J., Florence, P., Straub, J., Newcombe, R. & Lovegrove, S. DeepSDF: learning continuous signed distance functions for shape representation. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* (eds Gupta, A. et al.) 165–174 (IEEE, 2019).
44. Deng, C. et al. Vector neurons: a general framework for SO(3)-equivariant networks. In *Proc. IEEE/CVF International Conference on Computer Vision* (eds Hassner, T. et al.) 12180–12189 (IEEE, 2021).
45. Wang, Y. et al. Dynamic graph CNN for learning on point clouds. *ACM Trans. Graph.* **38**, 1–12 (2019).
46. Ho, J., Jain, A. & Abbeel, P. Denoising diffusion probabilistic models. In *Proc. 34th International Conference on Neural Information Processing Systems* (eds Larochelle, H. et al.) 6840–6851 (Curran Associates, 2020).
47. Kullback, S. & Leibler, R. A. On information and sufficiency. *Ann. Math. Stat.* **22**, 79–86 (1951).
48. Hoogeboom, E., Nielsen, D., Jaini, P., Forré, P. & Welling, M. Argmax flows and multinomial diffusion: learning categorical distributions. In *Proc. 35th International Conference on Neural Information Processing Systems* (eds Beygelzimer, A. et al.) 12454–12465 (Curran Associates, 2021).
49. Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
50. Jing, B., Eismann, S., Suriana, P., Townshend, R. J. L. & Dror, R. Learning from protein structure with geometric vector perceptrons. In *Proc. 11th International Conference on Learning Representations* (eds Nickel, M. et al.) (OpenReview.net, 2021).
51. Garcia Satorras, V., Hoogeboom, E., Fuchs, F., Posner, I. & Welling, M. E(n) equivariant normalizing flows. In *Proc. 35th International Conference on Neural Information Processing Systems* (eds Beygelzimer, A. et al.) 4181–4192 (Curran Associates, 2021).
52. Torge, J., Harris, C., Mathis, S. V. & Lio, P. DiffHopp: a graph diffusion model for novel drug design via scaffold hopping. In *Proc. ICML 2023 workshop on Computational Biology* (PMLR, 2023); https://icml-compbio.github.io/2023/papers/WCBICML2023_paper69.pdf
53. Dhariwal, P. & Nichol, A. Q. Diffusion models beat GANs on image synthesis. In *Proc. 35th International Conference on Neural Information Processing Systems* (eds Beygelzimer, A. et al.) 8780–8794 (Curran Associates, 2021).
54. Eberhardt, J., Santos-Martins, D., Tillack, A. F. & Forli, S. Autodock Vina 1.2.0: new docking methods, expanded force field, and Python bindings. *J. Chem. Inf. Model.* **61**, 3891–3898 (2021).

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Author contributions

X.N. conceived the research. X.N. obtained funding for the research. Z.C. and X.N. designed the research. Z.C. and X.N. conducted the research, including data curation, formal analysis, methodology design and implementation, result analysis and visualization. Z.C., B.P. and X.N. drafted the original paper. T.Z. provided comments on case studies for protein targets. D.A.-A. provided comments on case studies for low-quality examples. Z.C., B.P. and X.N. conducted the paper editing and revision. All authors reviewed the final paper.

Competing interests

The authors declare no competing interests.

Additional information

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